

Swiss Institute of Banking and Finance



University of St.Gallen

Case Study Session 1: Valuation – Consolidated Edison & Chemex AG

3,130/4,130 Corporate Finance

Key messages of this exercise

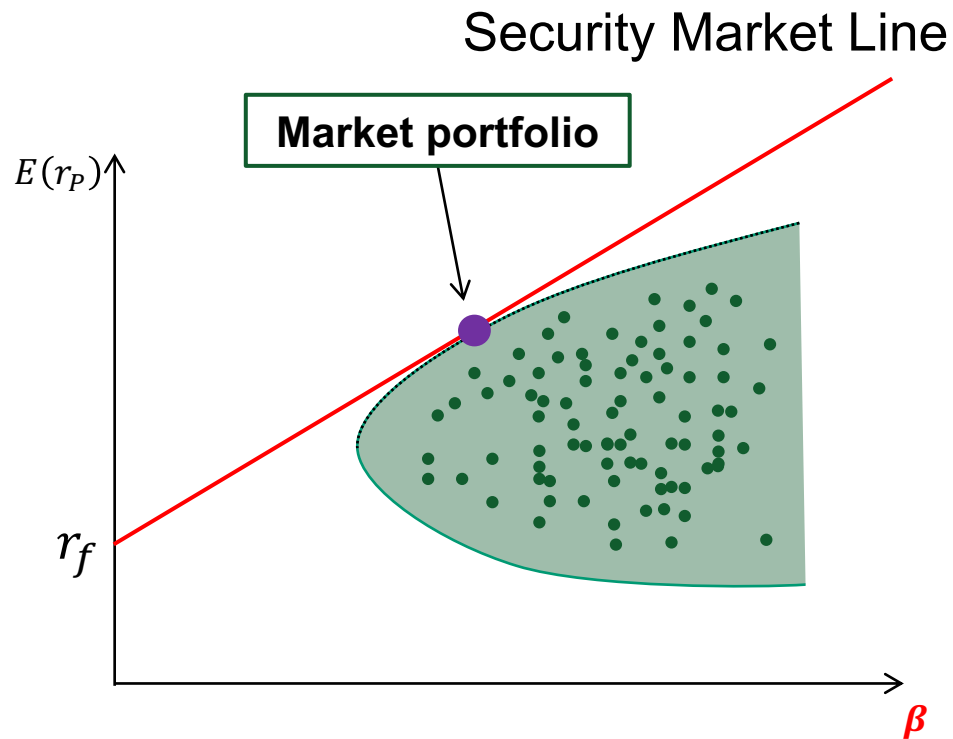
1. The Discounted Dividend Model and the Gordon Growth Model use dividends as valuation factors.
2. The discounted cash flow (DCF) method is based on free cash flows as the respective input factors.
3. Multiples value the firm based on peer comparison.

Outline

1. Recap: Portfolio theory
2. Consolidated Edison
3. Chemex AG

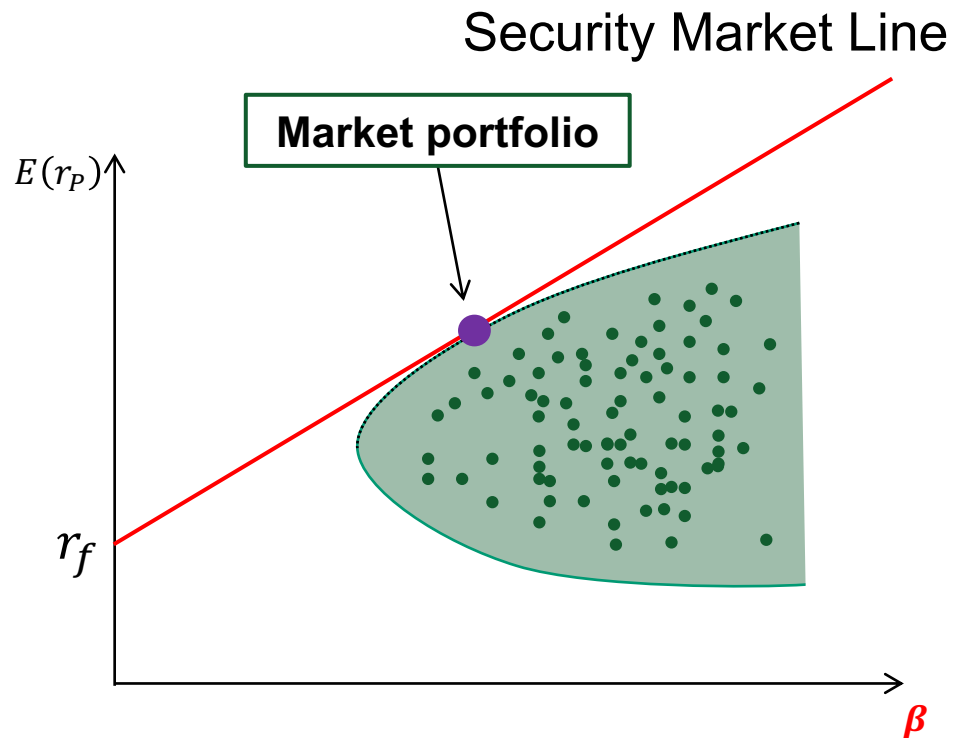
1. From the market portfolio to the CAPM

1. How are the market portfolio and beta related?



1. From the market portfolio to the CAPM

1. How are the market portfolio and beta related?



- Beta is a measure of the covariation of a particular asset with the market.
- Therefore, the beta of the market portfolio equals one ($\beta = 1$) since the covariation with itself is similar to its own variation.
- The variance of the risk-free asset is 0.
- (Almost) each risky asset is exposed to the market risk which can be measured by the beta.
- According to the CAPM, the expected return of a particular risky asset is proportional to its beta.

1. Capital Asset Pricing Model (CAPM)

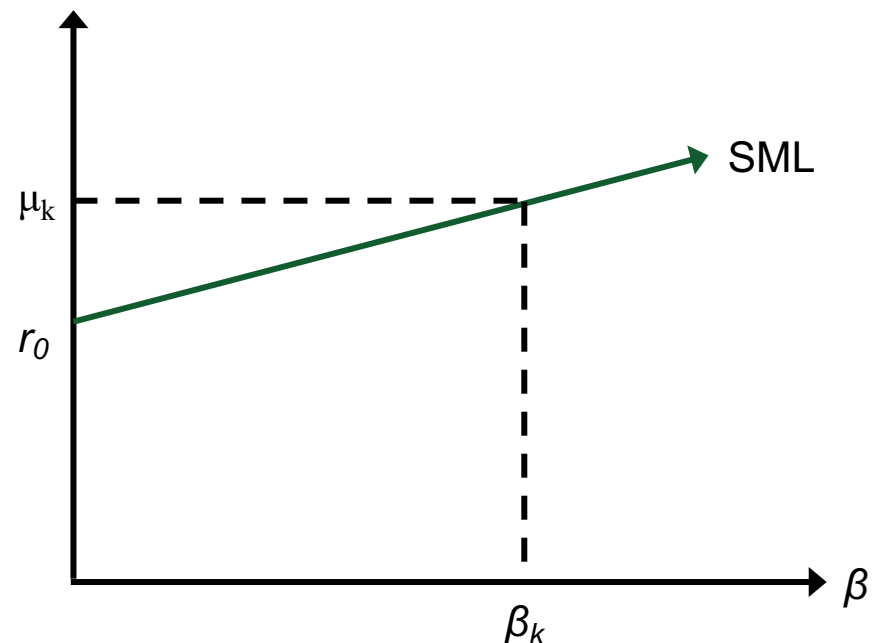
According to the CAPM, the expected return of every asset is determined (1) by the rate of interest, (2) by the systematic risk of the asset considered which is measured by beta (how the asset covaries with the market), and (3) the expected return of the market portfolio.

Formula:

$$\mu_k = r_0 + \beta_k \cdot (\mu_M - r_0)$$

Security-Market-Line (SML)

Expected return



1. Beta of an asset

1a. What is the expected price next year according to the CAPM?

Assumption:

- Current stock price (P_0) = CHF 50
- Next year dividend (D_1) = CHF 6
- Beta = 1.2
- Market risk premium = 10%
- Risk-free interest rate = 6%

1. Beta of an asset

1a. What is the expected price next year according to the CAPM?

Assumption:

- Current stock price (P_0) = CHF 50
- Next year dividend (D_1) = CHF 6
- Beta = 1.2
- Market risk premium = 10%
- Risk-free interest rate = 6%

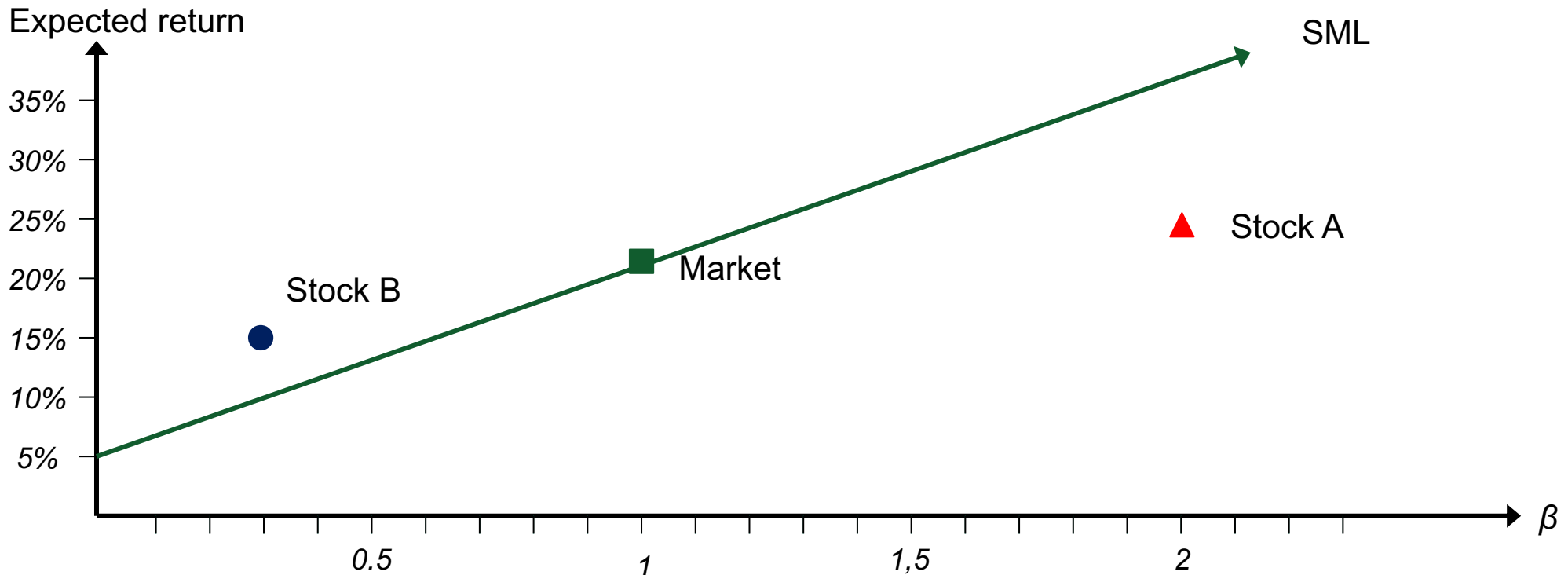
$$\begin{aligned}\mu_k &= r_0 + \beta_k \cdot (\mu_M - r_0) \\ &= 0.06 + 1.2 \cdot 0.1 = 0.18 = 18\%\end{aligned}$$

$$E(r_A) = \frac{P_1 + D_1 - P_0}{P_0} = \frac{P_1 + 6 - 50}{50} = 18\%$$

$$\rightarrow P_1 = \text{CHF } 53$$

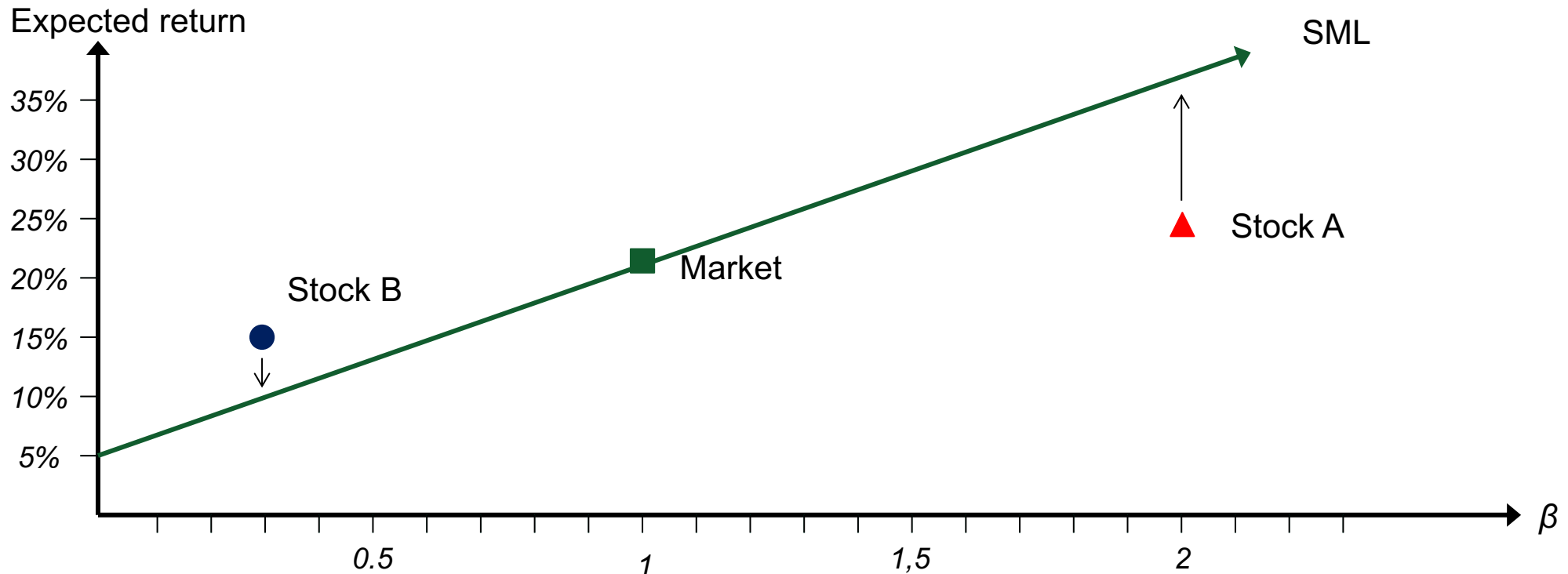
1. Security Market Line

1b. Suppose two stocks which lie above (Stock B) and below (Stock A) the Security Market Line (SML) respectively. What price reaction of the shares would you expect (ceteris paribus)?



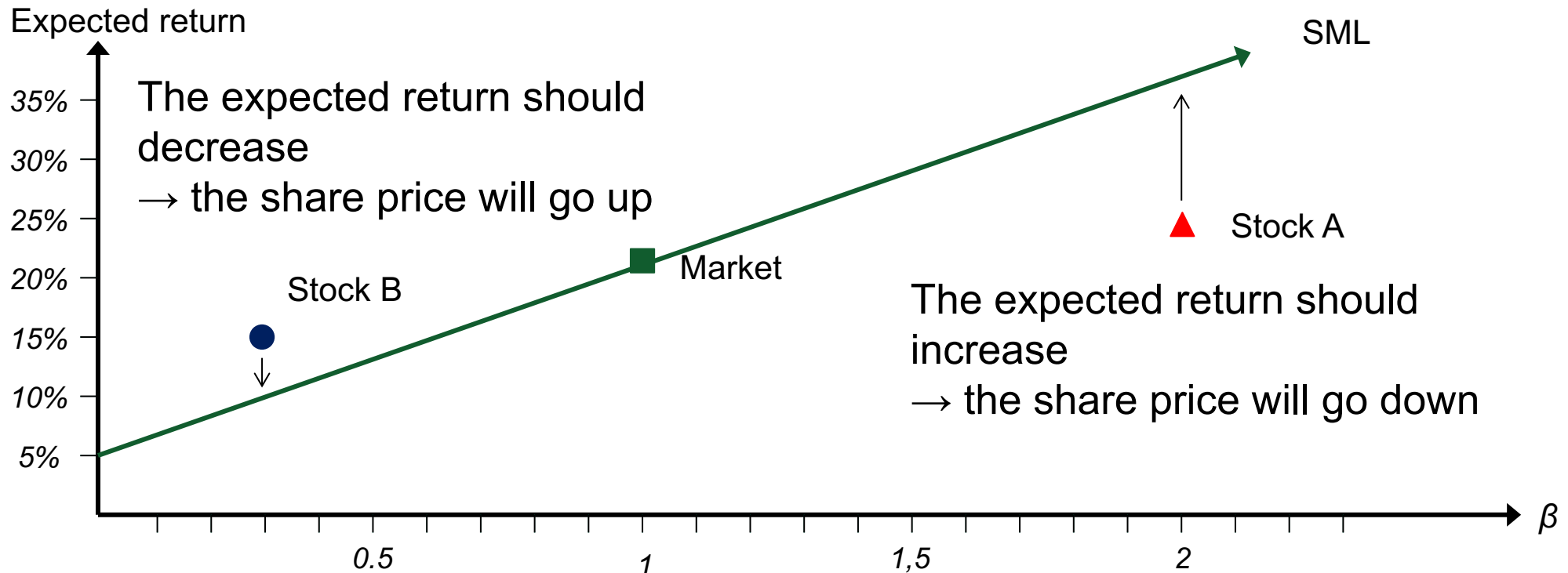
1. Security Market Line

1b. Suppose two stocks which lie above (Stock B) and below (Stock A) the Security Market Line (SML) respectively. What price reaction of the shares would you expect (*ceteris paribus*)?



1. Security Market Line

1b. Suppose two stocks which lie above (Stock B) and below (Stock A) the Security Market Line (SML) respectively. What price reaction of the shares would you expect (ceteris paribus)?



2. Valuation of Consolidated Edison

1. We will estimate the value of equity of Consolidated Edison applying the Dividend Discount Model.

$$P_0 = \frac{Div_1}{r}$$

2. In the next step, we will refine the model introducing the growth rate to estimate the value of equity with the Gordon Growth Model.

$$P_0 = \frac{Div_1}{r - g}$$

This case is based on the book “Investment Valuation” of Damodaran (3rd Edition).

2. Consolidated Edison – Corporate Profile



- For more than 180 years, Consolidated Edison, Inc. has served the world's most dynamic and demanding marketplace - metropolitan New York.
- Con Edison's principal business segments are Consolidated Edison Company of New York's regulated electric, gas and steam utility activities, Orange & Rockland Utilities' (O&R) regulated electric and gas utility activities, and Con Edison's competitive energy businesses.

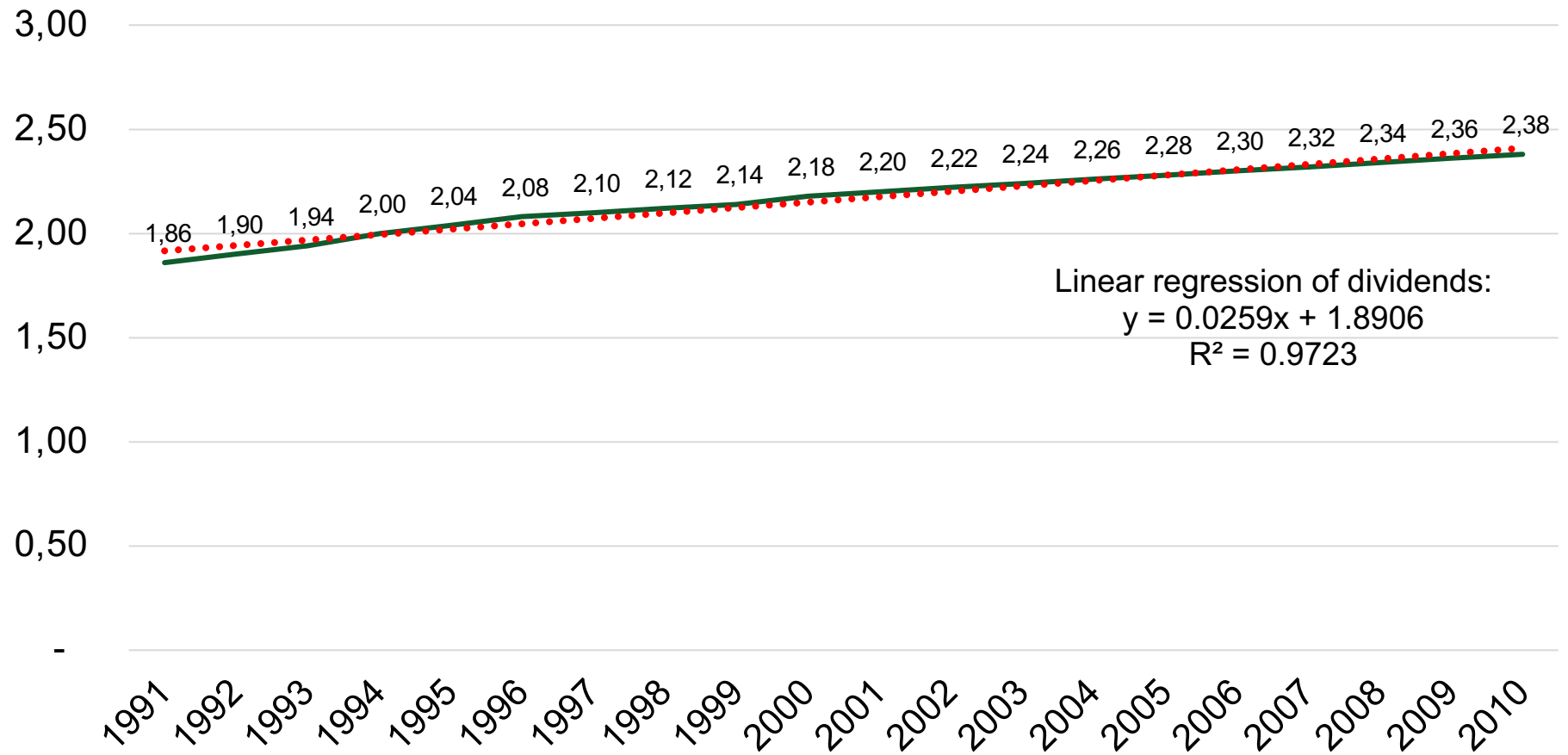
Source: "<https://www.conedison.com/en/about-us/our-businesses>"

2. Financial Highlights 2010

	(\$m)
Operating Revenues	13,325
Net Income	981
Total Assets	36,146
Market Capitalization	14,400
Capital Expenditure	2,029
Dividends per Share	\$2.38
Stock Price per Share (Year End)	\$49.57
5-year Annual Dividend Growth Rate	0.86%
Dividend payout ratio	69%

Source: Annual Report 2010 and Website

2. Yearly Dividend Payments per Share 1991-2010



Source: Consolidated Edison – Dividend History

2. Cost of Equity 2006-2010

2a. Calculate the cost of equity applying the CAPM!

Risk-free interest rate = 3.91%

Market risk premium = 4.60%

Beta = 0.25

2. Cost of Equity 2006-2010

2a. Calculate the cost of equity applying the CAPM!

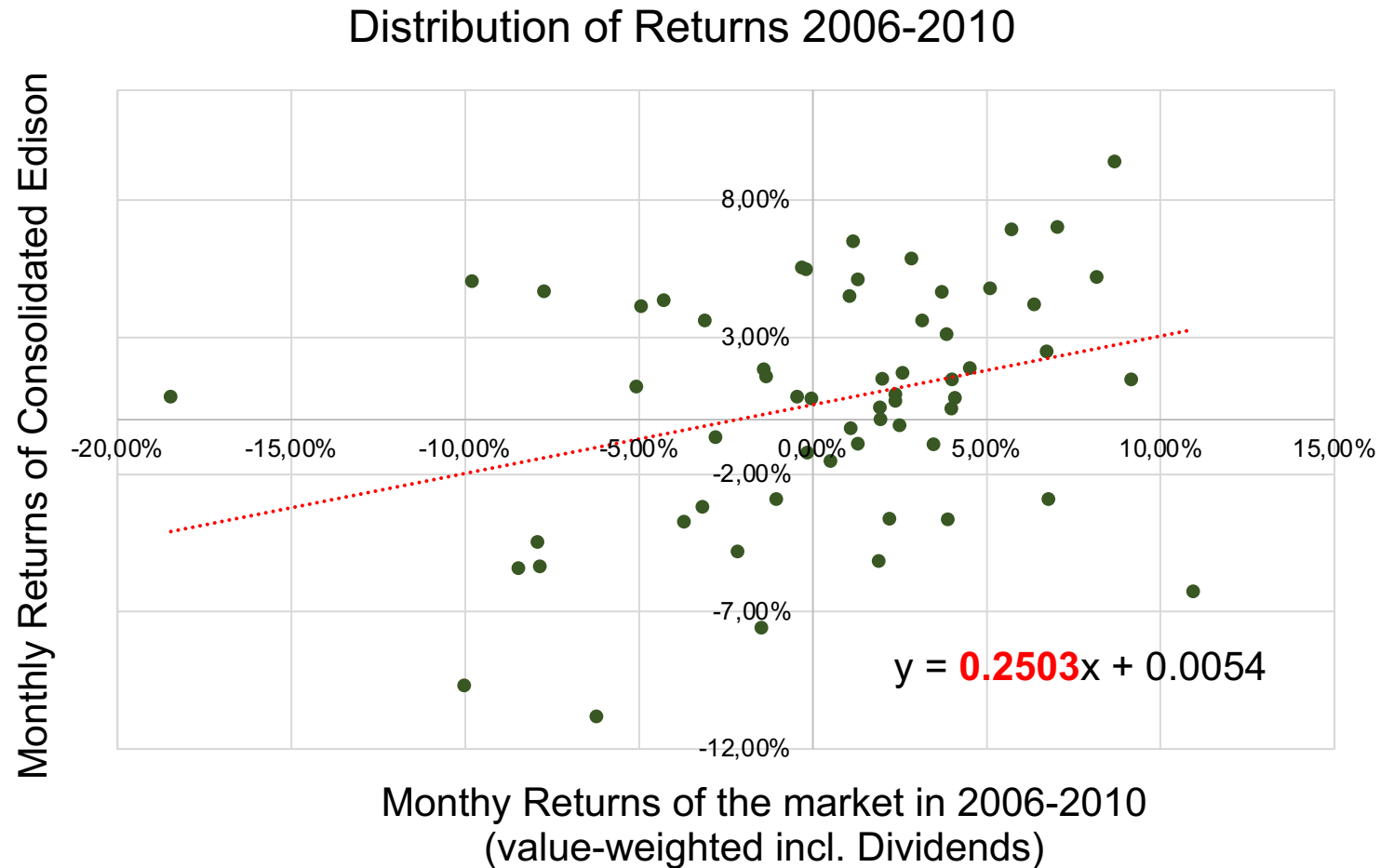
Risk-free interest rate = 3.91%

Market risk premium = 4.60%

Beta = 0.25

$$E(r_{Equ}) = 0.0391 + 0.25 \cdot 0.0460 = 0.0506 = 5.06\%$$

2. Intuition of the Beta 2006-2010



Source: CRSP

2. Dividend Discount Model 2011

2b. Calculate reasonable values for the expected next year dividend!

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Data from Financial Highlights:

$$Div_{t+1} = 2.38 \cdot (1 + 0.0086) = 2.4005 \approx 2.40$$

Linear prediction from historical dividends:

$$Div_{t+1} = 1.8906 + 0.0259 \cdot 20 = 2.4086 \approx 2.41$$

Growth rate from historical dividends:

$$r = \sqrt[19]{\frac{2.38}{1.86}} - 1 \approx 1.31\% \rightarrow Div_{t+1} = 2.38 \cdot (1 + 0.0131) = 2.4112 \approx 2.41$$

*19 years (1991-2010)

2. Dividend Discount Model 2011

2c. What would be the estimated stock price at the beginning of 2011 ?

2. Dividend Discount Model 2011

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Financial Highlights

$$P_0 = \frac{Div_1}{r} = \frac{2.40}{0.0506} = 47.43$$

Linear Predictions

$$P_0 = \frac{Div_1}{r} = \frac{2.41}{0.0506} = 47.63$$

Historical Growth Rate

$$P_0 = \frac{Div_1}{r} = \frac{2.41}{0.0506} = 47.63$$

**Cross-check: Actual
Dividend in 2011**

$$P_0 = \frac{Div_1}{r} = \frac{2.40^*}{0.0506} = 47.43$$

*(Financial Statement from 2011)

2. Gordon Growth Model 2011

2d. What would be the estimated stock price at the beginning of 2011 if you also consider the growth of the dividends?

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2d. What would be the estimated stock price at the beginning of 2011 if you also consider the growth of the dividends?

Financial Highlights

$$P_0 = \frac{Div_1}{r - g} = \frac{2.40}{0.0506 - 0.0086} = 57.14$$

Historical Growth Rate

$$P_0 = \frac{Div_1}{r - g} = \frac{2.41}{0.0506 - 0.0131} = 64.27$$

Stock price at the beginning of 2011

49.57

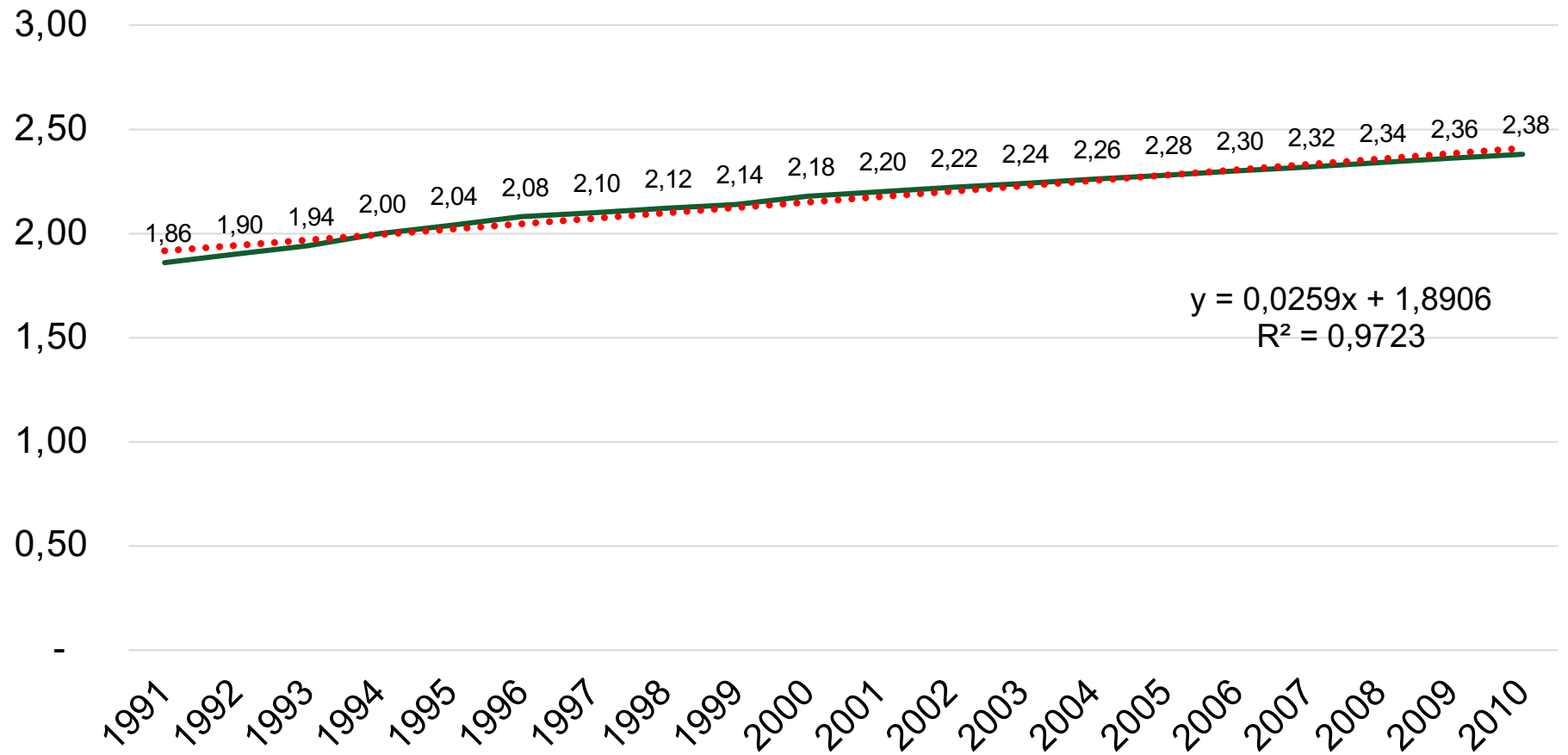
2. Gordon Growth Model

2e. Why does the Gordon Growth Model perform fairly well in this situation?

2. Background Information

- To understand why the Gordon Growth Model works well, we have to understand the firm and its environment.
- How did the dividends of Consolidated Edison grow in the past? Is a prediction of future dividends based on past dividends in this case reasonable? Does the payout ratio will change in the near future?
- Would it be problematic to determine the appropriate discount rate if Consolidated Edison has a volatile capital structure?
- What would be a good situation to estimate the growth rate? Does the environment of Consolidated Edison is similar to such an environment?

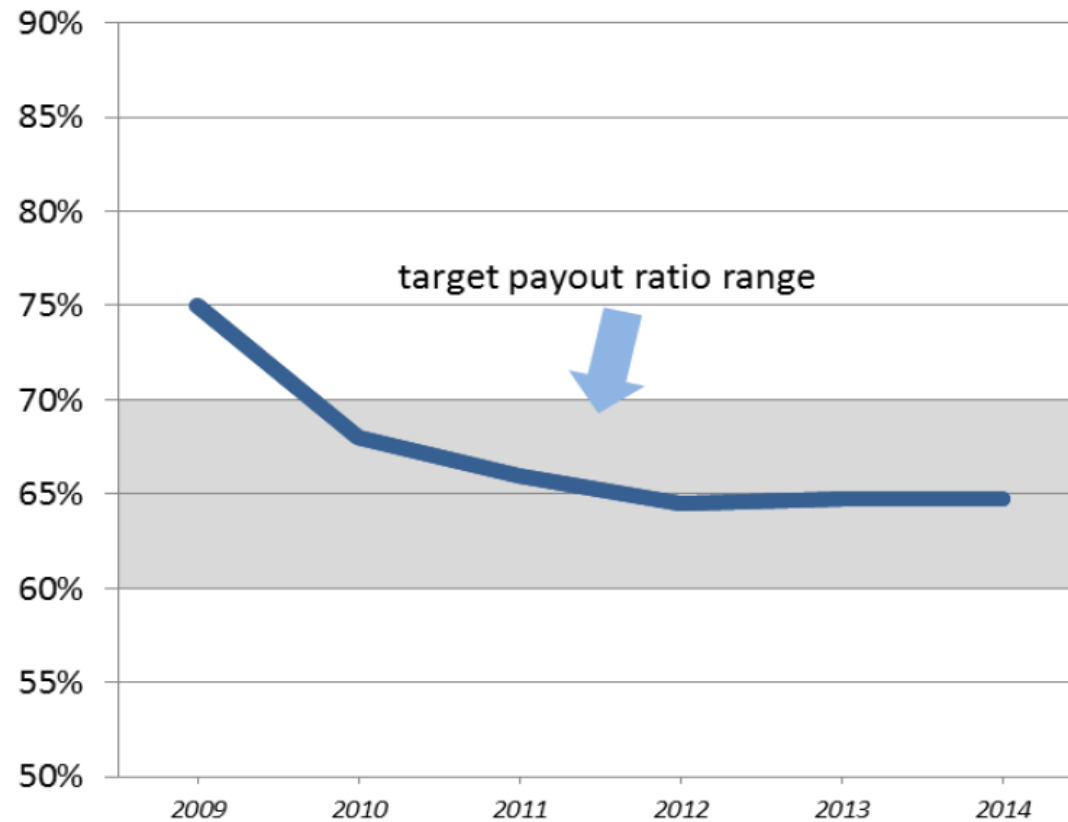
2. Yearly Dividend Payments per Share



Source: Consolidated Edison – Dividend History

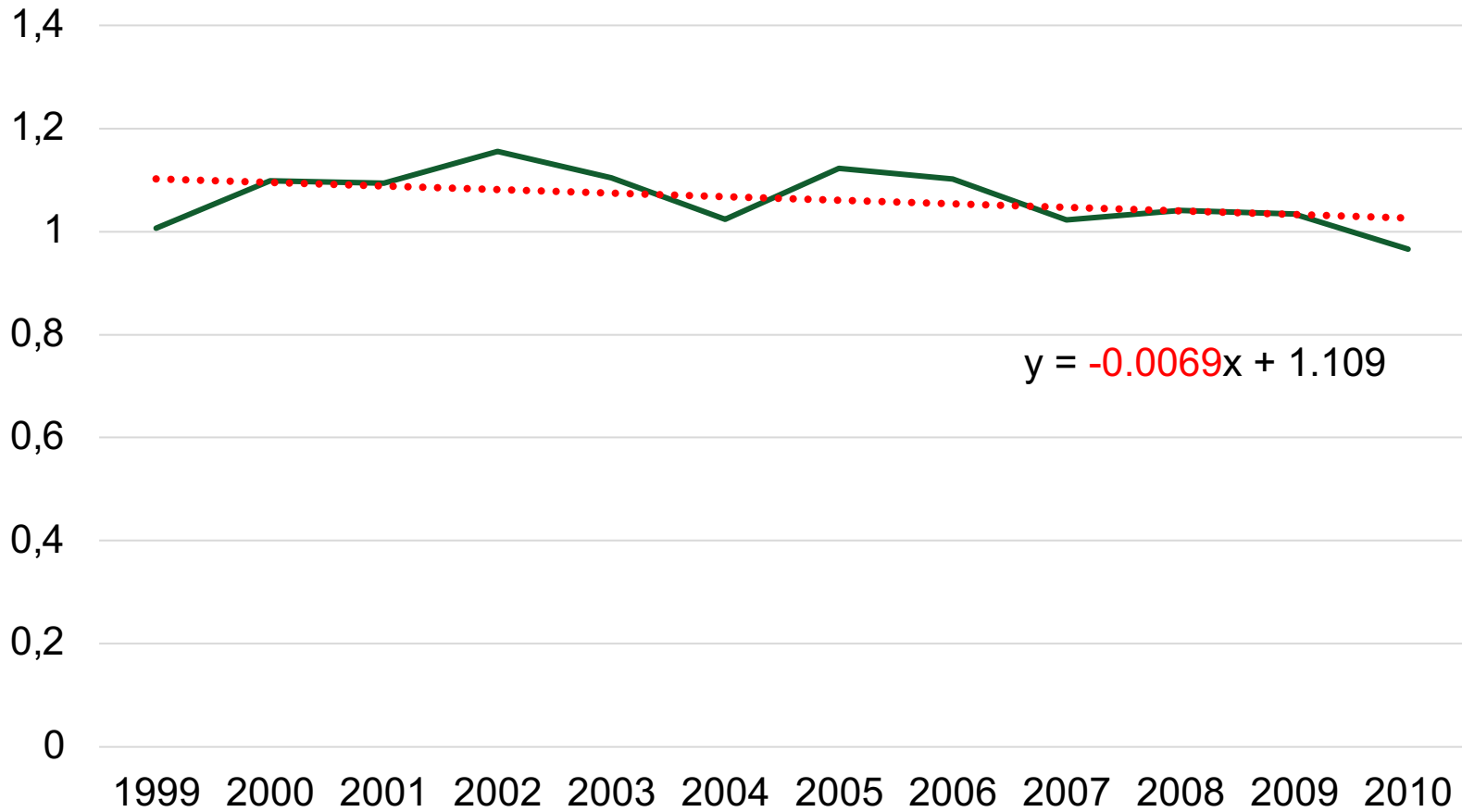
2. Dividend Payout Range

Dividend Payout Ratio 2009 – 2014



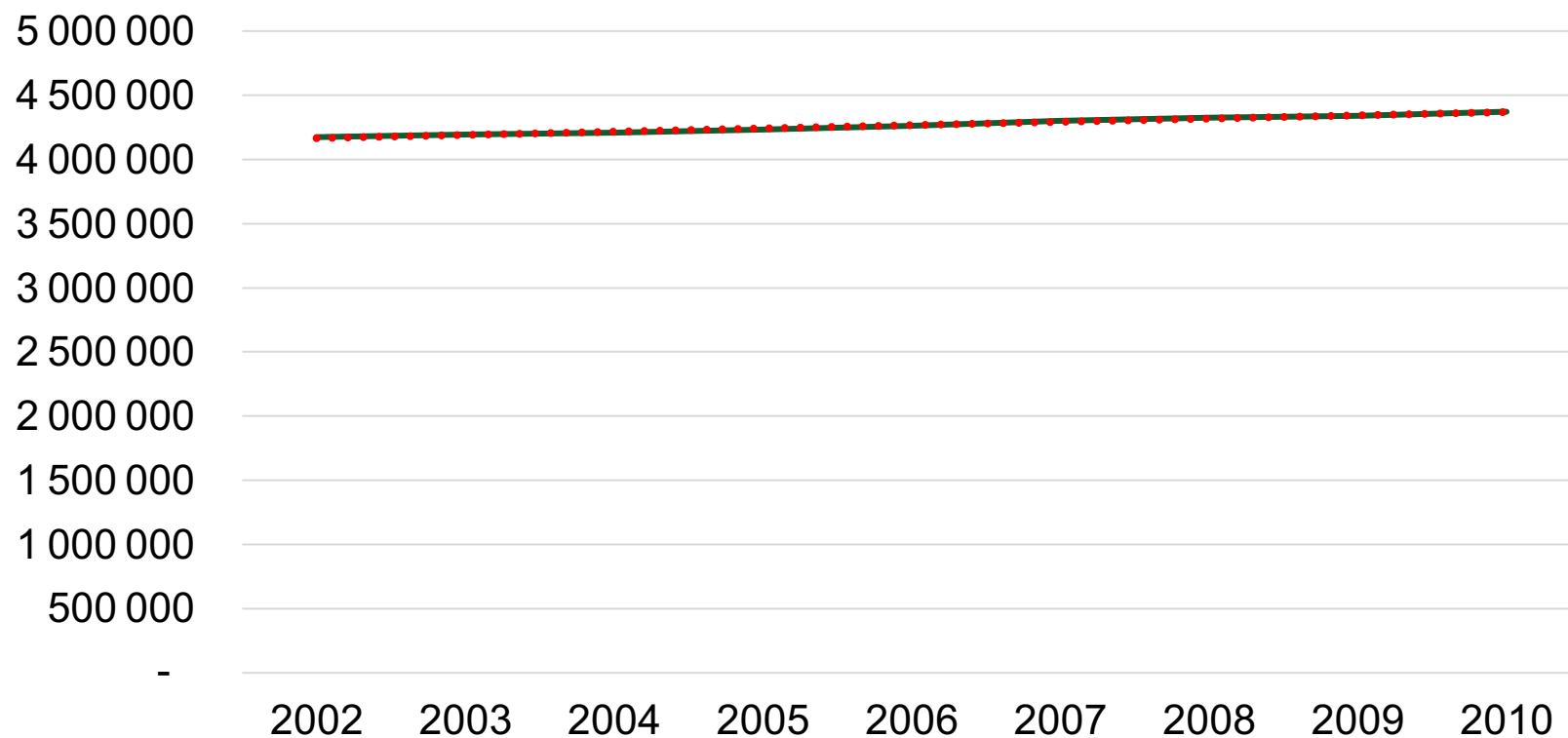
Source: 6-Year Financials and Operating Statistics 2009-2014

2. Capital Structure – Debt-to-Equity Ratio



Source: Compustat

2. Customers



Source: Consolidated Edison – Financial & Operating Statistics

2. Gordon Growth Model

2e. Why does the Gordon Growth Model perform fairly well in this situation?

- Since years, the environment of the firm (e.g. population and power usage) is quite stable.
- The firm is highly regulated and the prices for energy (e.g., electric, gas) and steam have to be about the inflation rate.
- The firm has a stable capital structure so that the cost of equity and the dividends remain stable.
- The firm's investors love dividends so that the firm pays out much of its earnings and free cash flows.

2. More recent past of Consolidated Edison

- We now look into more recent data to compute the price per share of Consolidated Edison using the Gordon Growth Model and test whether the model still holds.
- Does the Gordon Growth Model work well in the more recent past?
 - Are there any changes in the recent past which make it harder apply the Gordon Growth Model?
 - We will have a look into Consolidated Edison's dividend payments.
 - We will recalculate/ reestimate Consolidated Edison's cost of equity.

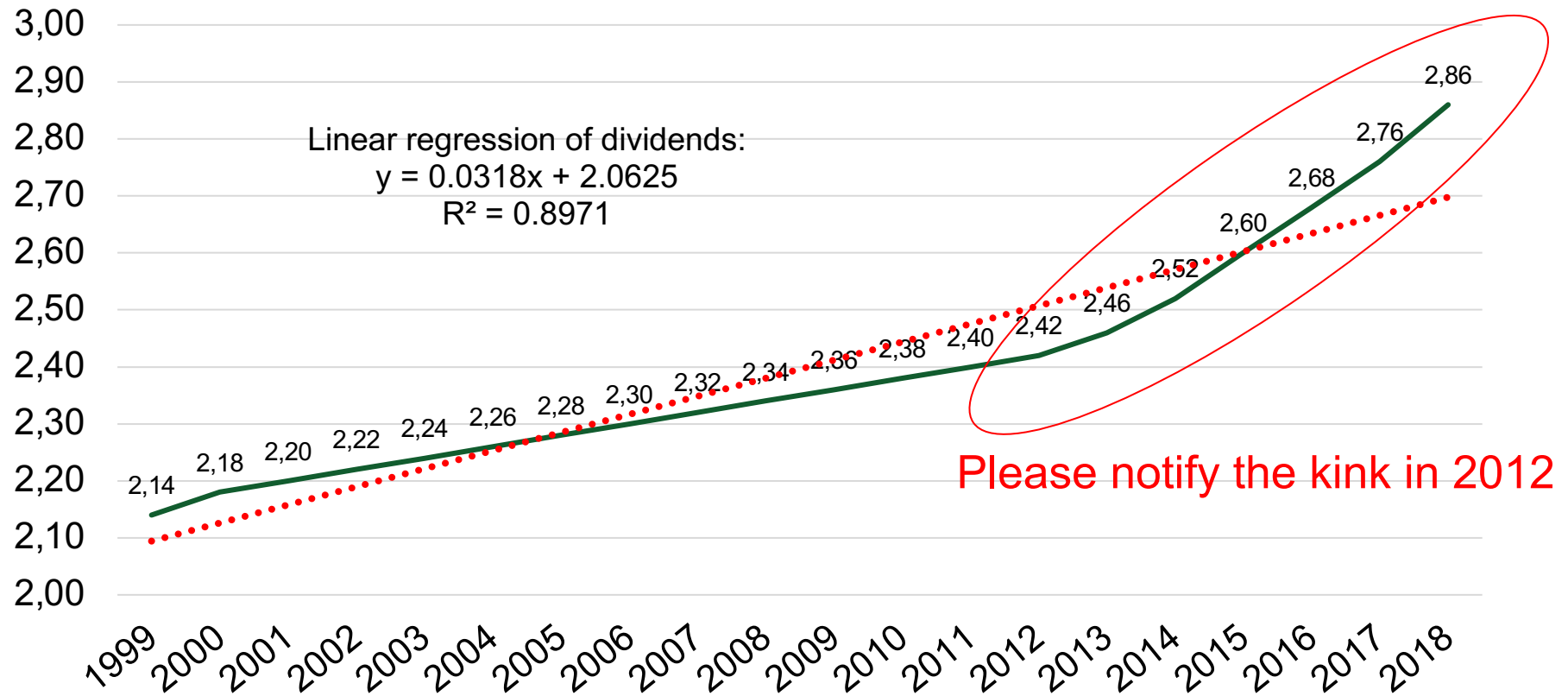
2. Financial Highlights 2018

	(\$m)
Operating Revenues	12,337
Net Income	1,382
Total Assets	53,920
Market Capitalization	24,500
Capital Expenditure	5,249
Dividends per Share	\$2.86
Stock Price per Share (Year End)	\$76.46
5-year Annual Dividend Growth Rate	3.06%
Dividend payout ratio	65%

vs. 0.86% in 2010

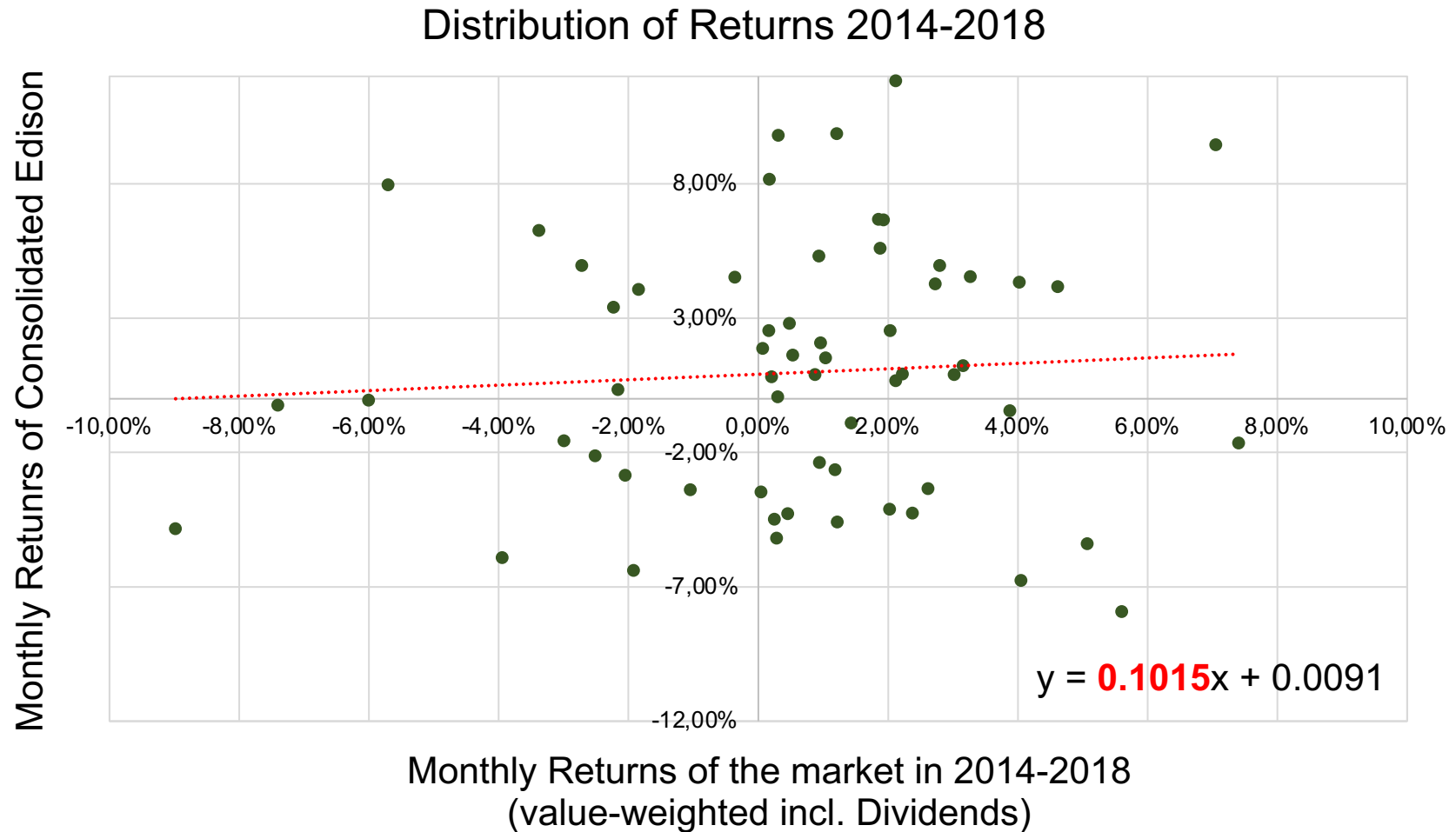
Source: Annual Report 2018 and Website

2. Yearly Dividend Payments per Share 1999-2018



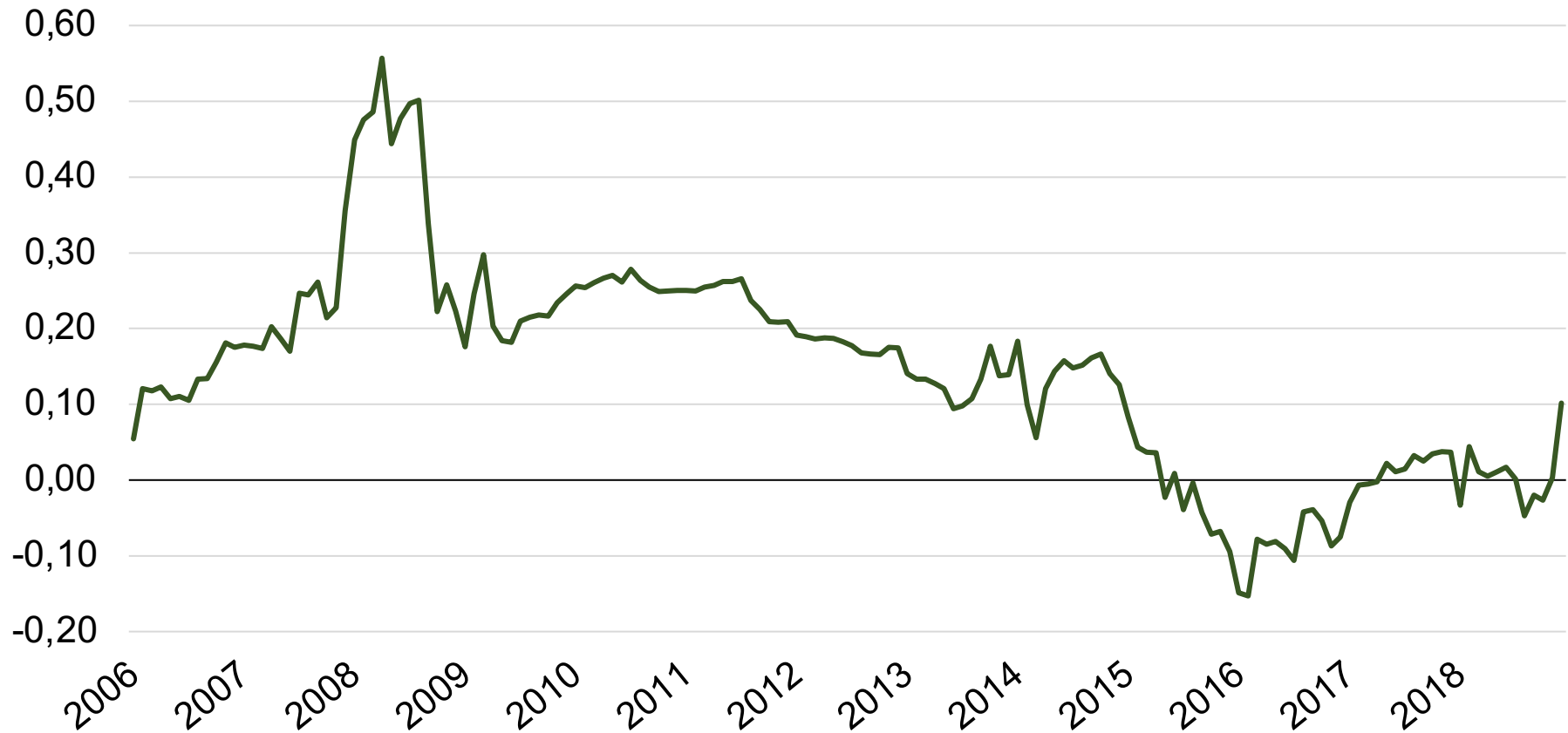
Source: Consolidated Edison – Dividend History

2. Intuition of the Beta 2014-2018



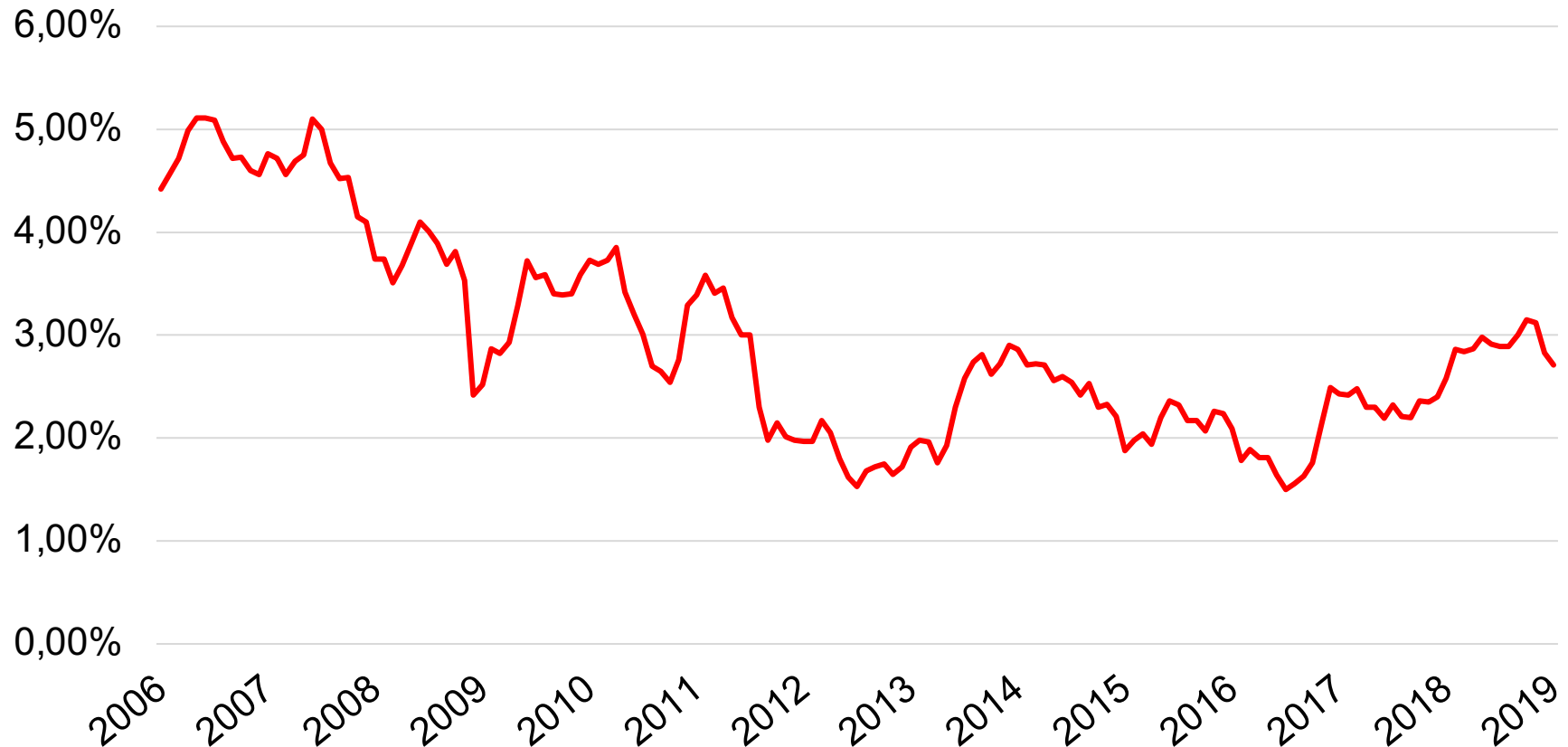
Source: CRSP

2. Five-year Rolling Monthly Beta 2006-2018



Source: CRSP

2. Monthly Risk-free Rate 2006-2018



Source: <https://fred.stlouisfed.org/series/GS10>

2. Cost of Equity 2014-2018

2f. Calculate the cost of equity applying the CAPM!

Risk-free interest rate = 2.35%

Market risk premium = 4.19%

Beta = 0.10

2. Cost of Equity 2014-2018

2f. Calculate the cost of equity applying the CAPM!

Risk-free interest rate = 2.35%

Market risk premium = 4.19%

Beta = 0.10

$$E(r_{Equ}) = 0.0235 + 0.10 \cdot 0.0419 = 0.02769 \approx 2.77\%$$

2. Dividend Discount Model 2019

2g. Calculate reasonable values for the expected next year dividend!

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Data from Financial Highlights:

$$Div_{t+1} = 2.86 \cdot (1 + 0.0306) = 2.9475 \approx 2.95$$

Linear prediction from historical dividends:

$$Div_{t+1} = 2.0625 + 0.0318 \cdot 20 = 2.6985 \approx 2.70$$

Growth rate from historical dividends:

$$r = \sqrt[19]{\frac{2.86}{2.14}} - 1 \approx 1.54\% \rightarrow Div_{t+1} = 2.86 \cdot (1 + 0.0154) = 2.9040 \approx 2.90$$

*19 years (1999-2018)

2. Gordon Growth Model 2019

2h. What would be the estimated stock price at the beginning of 2019 if you also consider the growth of the dividends?

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2h. What would be the estimated stock price at the beginning of 2019 if you also consider the growth of the dividends?

Financial Highlights

$$P_0 = \frac{Div_1}{r - g} = \frac{2.95}{0.0277 - 0.0306} = -1017.24$$

Historical Growth Rate

$$P_0 = \frac{Div_1}{r - g} = \frac{2.90}{0.0277 - 0.0154} = 235.77$$

Stock price at the beginning of 2019

75.14

2. Conclusion

- The Gordon Growth Model does not yield reasonable results in the more recent past.
- When we use data over the period 2014-2018, the predicted price per share for Consolidated Edison is negative because the cost of equity is lower than the dividend growth rate ($r < g$).
- This small and even negative denominator ($r - g$) is due to:
 - Higher growth dividend rate (g)
 - Lower cost of equity (r), itself due to decreasing beta and decreasing risk-free rate

2. Discounted Cash Flow (DCF) Model

- Instead of using dividends in the Dividend Discount Model (DDM), free cash flows (FCF) can also be used to **value a company**.
- The value of a business is usually computed as the discounted value of free cash flows out to a valuation horizon (H), plus the forecasted value of the business PV_H at this horizon, also discounted back to present value.

$$PV_0 = \frac{FCF_1}{1+r} + \frac{FCF_1}{(1+r)^2} + \dots + \frac{FCF_H}{(1+r)^H} + \frac{PV_H}{(1+r)^H} = \sum_{t=1}^H \frac{FCF_t}{(1+r)^t} + \frac{PV_H}{(1+r)^H}$$

Sum of present values of the
free cash flows planned **in detail**
for the coming H years

Discounted
horizon
value

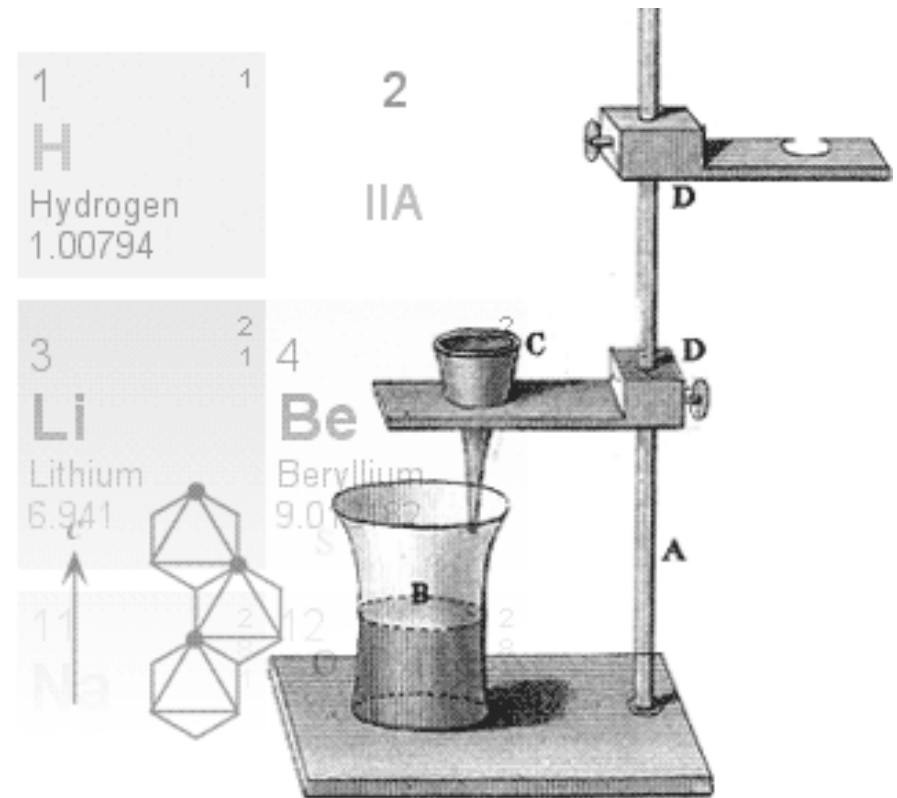
3. «Chemex AG»

Chemex AG is a leading international company in the specialty chemicals sector with a strong customer focus and innovative products.

Its market segments include:

- Textiles, leather & paper chemicals
- Pigments and chem. additives
- Monomers and water-soluble polymers
- Life-science chemicals

A private equity firm is attracted by the long-term operational performance and thus interested in buying the Chemex AG.



3. Condensed Annual Report 2018

Balance Sheet 2018

Assets	(\$m)
Cash (50% excess liquidity)	550
Receivables	1'100
Inventories	1'300
Plan, property, equip. (PPE)	2'400
	5'350
Liabilities	(\$m)
Accounts payable	1'200
Bank loans	50
Long-term debt	2'100
Equity* (incl. net income – dividend)	2'000
	5'350

Income Statement 2018

Sales	8'500
- Cost of sales	-7'600
EBITDA	900
<i>Earnings before interests, taxes, depreciation and amortization</i>	
- Depreciation	-340
EBIT	560
<i>Earnings before interests and taxes</i>	
- Interest expenses	-127
EBT	433
<i>Earnings before taxes</i>	
- Taxes	-108
Earnings after taxes/ Net income	325

*100m shares outstanding

3. Forecast Investment Plan

Assume that payments for investments occur at the end of each year.

	2019	2020	2021	2022	2023	2024
Plant, property & equip.	2'400	2'412	2'453	2'509	2'538	2'558
CAPEX	300	330	350	330	325	= depreciation
Depreciation	-288	-289	-294	-301	-305	-307
Production equipment	2'412	2'453	2'509	2'538	2'558	2'558

3. Forecast Income Statement

	%	2019	2020	2021	2022	2023	2024
Sales		8'755	9'018	9'198	9'382	9'476	9'571
<i>Growth forecast in sales</i>		3%	3%	2%	2%	1%	1%
- Cost of goods sold		-7'880	-8'116	-8'278	-8'444	-8'528	-8'614
EBITDA		875	902	920	938	948	957
- Depreciation		-288	-289	-294	-301	-305	-307
EBIT		587	613	626	637	643	650
- Interests		-130	-133	-135	-137	-139	-142
EBT		457	480	491	500	504	508
- Tax	25%	-114	-120	-123	-125	-126	-127
Net income		343	360	368	375	378	381
Dividend	90%	309	324	331	338	340	343

3. Forecast Balance Sheet

Assets (in \$ million)	2019	2020	2021	2022	2023	2024
Cash & Equivalents	424	418	407	430	473	537
Accounts receivable	1'151	1'186	1'210	1'234	1'246	1'259
Inventories	1'313	1'353	1'380	1'407	1'421	1'436
Plant, properties & equip.	2'412	2'453	2'509	2'538	2'558	2'558
<i>(CAPEX as part of PPE)</i>	<i>(300)</i>	<i>(330)</i>	<i>(350)</i>	<i>(330)</i>	<i>(325)</i>	<i>(307)</i>
Total Assets	5'300	5'410	5'506	5'609	5'698	5'790
Liabilities (in \$ million)	2019	2020	2021	2022	2023	2024
Accounts payable	1'079	1'112	1'134	1'157	1'168	1'180
Short term debt	37	27	15	32	47	38
Long term debt	2'150	2'200	2'250	2'275	2'300	2'350
Equity	2'034	2'071	2'107	2'145	2'183	2'222
Total Liabilities & Equity	5'300	5'410	5'506	5'609	5'698	5'790

3. Free Cash Flows

Free cash flow:

Cash flows created by the firm which are available for shareholders and creditors.

Free cash flow calculation

EBIT

- Taxes on EBIT

NOPAT

Net operating profit after taxes

+ Depreciation

Cash flow

- *increase of working capital*

- CAPEX

Free cash flow

3. Forecast the free cash flows

	2019	2020	2021	2022	2023	2024
EBIT	587	613	626	637		
- Taxes	-147	-153	-157	-159		
NOPAT	440	460	469	478		
+ Depreciation	288	289	294	301		
Change in working capital	-185	-42	-29	-28		
CAPEX	-300	-330	-350	-330		
Free cash flow	243	377	384	421		
Calculate the change in working capital						
Increase in receivables	51	35	24	24		
Increase in inventory	13	40	27	27		
- Increase in payables	121	-33	-22	-23		
Increase in working capital	185	42	29	28		

3. Forecast the free cash flows

	2019	2020	2021	2022	2023	2024
EBIT	587	613	626	637	643	650
- Taxes	-147	-153	-157	-159	-161	-163
NOPAT	440	460	469	478	482	487
+ Depreciation	288	289	294	301	305	307
Change in working capital	-185	-42	-29	-28	-15	-16
CAPEX	-300	-330	-350	-330	-325	-307
Free cash flow	243	377	384	421	447	471
Calculate the change in working capital						
Increase in receivables	51	35	24	24	12	13
Increase in inventory	13	40	27	27	14	15
- Increase in payables	121	-33	-22	-23	-11	-12
Increase in working capital	185	42	29	28	15	16

3. Parts of the company value

Company value		
Forecasting horizon		Horizon value
Sum of the discounted free cash flows over the forecasting horizon		The 'horizon value' is a lump-sum estimation of the <i>present value at a future point in time</i> of all subsequent future cash flows (typically assumed to grow perpetually at a constant rate). $\text{Horizon value}_T = \frac{FCF_{T+1}}{\text{capital cost} - g}$

3. Forecast the horizon value

Calculate the horizon value of Chemex AG

Assumptions:

Cost of capital = 8.5%

Organic growth rate = 2.0%

Horizon value₂₀₂₄ =

3. Forecast the horizon value

Calculate the horizon value of Chemex AG

Assumptions:

Cost of capital = 8.5%

Organic growth rate = 2.0%

free cash flow₂₀₂₄ = 471

$$\text{Horizon value}_{2024} = \frac{FCF_{2025}}{r - g} = \frac{471 \cdot (1 + 0.02)}{0.085 - 0.02} = 7'391$$

3. Sensitivity analysis of the horizon value

		Cost of Capital								
		7.70%	7.90%	8.10%	8.30%	8.50%	8.70%	8.90%	9.10%	9.30%
Terminal growth rate	1.2%	7'333	7'114	6'908	6'713	6'529	6'355	6'190	6'034	5'885
	1.4%	7'581	7'348	7'128	6'922	6'727	6'542	6'368	6'203	6'045
	1.6%	7'845	7'596	7'362	7'142	6'935	6'740	6'555	6'380	6'215
	1.8%	8'127	7'860	7'611	7'377	7'156	6'949	6'753	6'568	6'393
	2.0%	8'428	8'143	7'876	7'626	7'391	7'170	6'963	6'766	6'581
	2.2%	8'752	8'445	8'159	7'891	7'641	7'406	7'185	6'976	6'780
	2.4%	9'100	8'769	8'461	8'175	7'907	7'656	7'420	7'199	6'990
	2.6%	9'475	9'118	8'786	8'478	8'191	7'922	7'671	7'435	7'213
	2.8%	9'881	9'494	9'136	8'803	8'495	8'207	7'938	7'686	7'449

3. Appropriate growth rate

- ✓ The assumed growth rate g has a large impact on the company value.
 - Few companies can grow faster than the broad economy forever.
 - A good estimate is the sum of (1) the **long-term growth rate** of the respective market volume and (2) the **expected inflation rate**.

3. Enterprise and equity value

Calculate the enterprise and equity value!

Operating result (PV of FCF)

- + excess liquidity
- + investment in unconsolidated subsidiaries
- + other non-operative asset values

Enterprise value

- interest bearing debt
- capitalized value of operative leasing
- minority interests
- preferred stocks
- employee stock options

Equity value

Note: Preferred stocks are treated as equity in other legal systems (e.g. Germany).

3. Enterprise and equity value

DCF value of the company	2019	2020	2021	2022	2023	2024
Free cash flows						
Horizon value (g = 2%)						
Discount factor (WACC = 8.5%)						
Discounted free cash flows						
Discounted horizon value						

Sum of discounted free cash flows	
Discounted horizon value	
Operative value of the company	
+ excess liquidity	
Enterprise value	
- interest bearing debt	
Equity value	

3. Enterprise and equity value

DCF value of the company	2019	2020	2021	2022	2023	2024
Free cash flows	243	377	384	421	447	471
Horizon value (g = 2%)						7391.1
Discount factor (WACC = 8.5%)	0.9217	0.8495	0.7829	0.7216	0.6650	0.6129
Discounted free cash flows	224.0	320.3	300.6	303.8	297.3	288.7
Discounted horizon value						4'530.0

Sum of discounted free cash flows	1'734.7
Discounted horizon value	4'530.0
Operative value of the company	6'264.7
+ excess liquidity (50% of 550)	+ 275
Enterprise value	6'539.7
- interest bearing debt	-2'150
Equity value	4'389.7

3. Impact of CAPEX

How does the value of the enterprise and the equity change if the CAPX will be reduced by 50 every year?

3. Enterprise and equity value

DCF value of the company	2019	2020	2021	2022	2023	2024
Free cash flows						
Change in CAPEX						
Horizon value (g = 2%)						
Discount factor (WACC = 8.5%)	0.9217	0.8495	0.7829	0.7216	0.6650	0.6129
Discounted free cash flows						
Discounted horizon value						

Sum of discounted free cash flows	
Discounted horizon value	
Operative value of the company	
+ excess liquidity	
Enterprise value	
- interest bearing debt	
Equity value	

3. Enterprise and equity value

DCF value of the company	2019	2020	2021	2022	2023	2024
Free cash flows	243	377	384	421	447	471
Change in CAPEX	50	50	50	50	50	50
Horizon value (g = 2%)						8'175.7
Discount factor (WACC = 8.5%)	0.9217	0.8495	0.7829	0.7216	0.6650	0.6129
Discounted free cash flows	270.1	362.7	339.8	339.9	330.5	319.3
Discounted horizon value						5'011.1

Sum of discounted free cash flows	1'962.3
Discounted horizon value	5'011.1
Operative value of the company	6'973.4
+ excess liquidity	+ 275
Enterprise value	7'248.4
- interest bearing debt	-2'150
Equity value	5'098.4

3. Interpretation of the results

Do lower CAPEX really increases the firm value?

3. Interpretation of the results

Do lower CAPEX really increases the firm value?

- CAPEX is the yearly investment in fixed (or intangible) assets to maintain or grow them.
- Suppose the management would cut CAPEX completely. As a consequence, the firm's asset base would “dry out” so that the cash flows from the firm's assets would decrease.
- Therefore, CAPEX and cash flows depend on each other.
- However, the example shows that a slightly forecasted decrease of CAPEX increases the calculated firm value (if all other factors remain constant)
- So, the forecasted cash flows have to be consistent with the forecasted CAPEX.

3. Evaluation of DCF

Assess the advantages and drawbacks of the DCF valuation method!

3. Advantages of DCF



3. Advantages of DCF

- + Allows flexible growth projections over forecasting horizon (i.e. when forecasting accuracy is highest)
- + Supports value-oriented evaluation of investment strategies
- + Considers operational initiatives through planned financial statements
- + Based on cash flows, which are less affected by the accounting standards than earnings
- + Is relatively independent of market fluctuations and control premium
- + Founded on a theoretical valuation concept

3. Disadvantages of DCF



3. Disadvantages of DCF

- Results very sensitive to the assumption of cost of capital, growth expectations, and forecasted cash flows.
- Only a small part of the company value is typically covered by the forecast period (compared to horizon value).
- Insufficient consideration of the flexibility of operations, the synergy effect and control premium.

3. Multiple Valuation

How does the multiple valuation method work?

What are advantages and drawbacks?

3. Multiple Valuation

Multiple valuation

- **Concept:**
- **Construction:**
- **Categories:**
- **Approach:**
 - 1.
 - 2.
 - 3.
 - 4.
 - 5.

3. Multiple Valuation

Multiple valuation

- **Concept:** Calculate an asset value by multiplying (1) the value per unit (i.e. the multiple) with (2) the number of units.
- **Construction:** A multiple relates (1) an asset value (e.g. the enterprise value) to (2) a fundamental value driver (income, cash flow, or book value)
- **Categories:** (1) Enterprise Value (EV) – value driver before interest expense, e.g. EV/EBIT(DA) ratio, EV/Total Assets ratio; (2) Equity Value – value driver after interest expense, e.g. Price/Earnings ratio, Market/Book ratio, etc.
- **Approach:**
 1. Identify peer group
 2. Select multiple
 3. Calculate the multiple for each member of the peer group (and adjust for company-specific differences)
 4. Take the median / average
 5. Multiply the resulting multiple with the value driver of the evaluated company

3. Multiple Valuation

Multiple valuation	
Advantages	Drawbacks
Hints	

3. Multiple Valuation

Multiple valuation

Advantages

- Incorporates capital market information
- Allows benchmark a company to the market

Drawbacks

- Difficult to account for company-specific differences, such as growth prospects, risk exposure, leverage, company size, etc.

Hints

- Empirically, multiple valuation provides similar accuracy as the DCF method
- Multiple valuation based on forecasted value driver improves accuracy

3. Multiple Valuation

Suppose the following peers:

Peer companies	Last Price	P/E Forecast FYR1	Exp. growth rate
TeChem Ltd.	32.41	10.9	1.9%
Chem Specialty SA	49.04	13.2	2.2%
Chemical & Brothers AG	11.03	14.1	2.1%
SynChem, Inc.	547.25	9.4	1.2%
CSP Co	48.52	9.8	1.3%
Sinochemicals, Inc.	101.80	10.6	1.8%
AirChemicals Co	13.91	12.5	2.0%
Average		11.5	1.8%
Median		10.9	1.9%

3. Multiple Valuation

Please estimate the value of Chemex AG using the P/E ratios!

P/E ratio

3. Multiple Valuation

Please estimate the value of Chemex AG using the P/E ratios!

P/E ratio

The estimated earnings of the Chemex AG are USD 343m in 2019.
100m shares are outstanding.

Average

$$11.5 = \frac{MV(Equity)}{343}$$

$$MV(Equity) = 3944.5$$

Median

$$10.9 = \frac{MV(Equity)}{343}$$

$$MV(Equity) = 3738.7$$

3. Multiple Valuation

What is the P/E ratio considering the DCF valuation?
Is the company over- or undervalued compared with its peers?

Implied P/E ratio

3. Multiple Valuation

What is the P/E ratio considering the DCF valuation?

Is the company over- or undervalued compared with its peers?

Implied P/E ratio

The estimated earnings of the Chemex AG are USD 343m in 2019.

100m shares are outstanding.

Value of equity taken from the DCF is 4'389.7.

$$\frac{P}{E_{impl}} = \frac{Equity_{DCF}}{Earnings_{t+1}} = \frac{4389.7}{343} = 12.8$$

Chemex seems to be more valuable than its peers. However, the different growth rates are not considered.

3. Multiple Valuation

Is Chemex over- or undervalued if the growth rates are considered?

Peer companies	P/E	g	PEG*
TeChem Ltd.	10.9	1.9%	
Chem Specialty SA	13.2	2.2%	
Chemical & Brothers AG	14.1	2.1%	
SynChem, Inc.	9.4	1.2%	
CSP Co	9.8	1.3%	
Sinochemicals, Inc.	10.6	1.8%	
AirChemicals Co	12.5	2.0%	
Average	11.5	1.8%	
Median	10.9	1.9%	

* growth-adjusted P/E

3. Multiple Valuation

Is Chemex over- or undervalued if the growth rates are considered?

Peer companies	P/E	g	PEG*
TeChem Ltd.	10.9	1.9%	5.7
Chem Specialty SA	13.2	2.2%	6.0
Chemical & Brothers AG	14.1	2.1%	6.7
SynChem, Inc.	9.4	1.2%	7.8
CSP Co	9.8	1.3%	7.5
Sinochemicals, Inc.	10.6	1.8%	5.9
AirChemicals Co	12.5	2.0%	6.3
Average	11.5	1.8%	6.6
Median	10.9	1.9%	6.3

growth-adjusted P/E

$$\frac{P/E}{g \cdot 100} = PEG$$

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3. Multiple Valuation

Is Chemex over- or undervalued if the growth rates are considered?

PEG ratio of Chemex

3. Multiple Valuation

Is Chemex over- or undervalued if the growth rates are considered?

PEG ratio of Chemex

$$\frac{P}{E_{impl}} = \frac{Equity_{DCF}}{Earnings_{t+1}} = \frac{4389.7}{343} = 12.8$$

Organic growth rate was 2%.

$$PEG_{impl} = \frac{12.8}{2} = 6.4 \text{ vs. } 6.6$$

Chemex' PE ratio is slightly higher than the average PE ratio but taking growth rates into account yields reasonable values. The mean and median PEG are 6.6 and 6.3, respectively which are close to Chemex' PEG of 6.4.

Recap

- The Dividend Discount Model and the Gordon Growth Model may perform well if the environment of the firm and the firm itself are relatively stable.
- The Discounted Cash Flow Model is based on the Free Cash Flows of a firm so that it is independent of any accounting or dividend policies. However, the model is highly sensitive to the several estimation parameters (e.g. terminal growth).
- In contrast to the DDM, GGM, and the DCF, multiple valuation uses comparable firms or transactions to estimate the value. Therefore, the “market mood” will be reflected.

Literature

Additional literature

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